

7 How can the acceleration of an object be determined?

- A from the area under a displacement–time graph
- B from the area under a velocity–time graph
- C from the gradient of a displacement–time graph
- D from the gradient of a velocity–time graph

8 The acceleration of free fall on the Earth is different to the acceleration of free fall on the Moon.

How do the mass and weight of an object on the Earth compare to its mass and weight on the Moon?

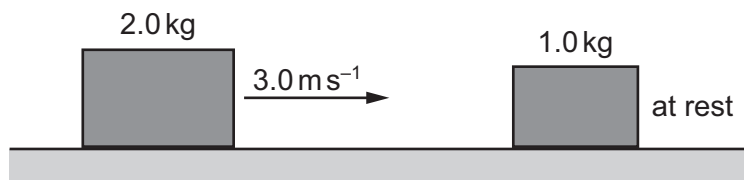
	mass	weight
A	different	different
B	different	same
C	same	different
D	same	same

9 A sphere moves vertically downwards at terminal (constant) velocity in a liquid.

Which statement about the magnitude of the upthrust acting on the sphere is correct?

- A It is proportional to the acceleration of free fall.
- B It is equal to the weight of the sphere.
- C It is proportional to the density of the sphere.
- D It is proportional to the square of the radius of the sphere.

10 An object of mass 2.0 kg is travelling at a speed of 3.0 ms^{-1} on a horizontal frictionless surface. This object collides head-on with a stationary object of mass 1.0 kg . The two objects stick together on impact.



How much kinetic energy is lost on impact?

- A zero
- B 2.0 J
- C 2.4 J
- D 3.0 J